

increased by nearly 9 per cent. The turnover of these producers had increased to a slightly greater extent than had that of members of the West Association between 1950 and 1953. The West Association had by 1950 been established over ten years, and the profit rates of its members only increased by just over 3 per cent. on sales between 1950 and 1953.

242. We have referred in paragraphs 224 and 225 to the trading figures of non-Association producers, including their trade in unwashed sand and gravel. We find that the rates of profit made on the unwashed products, according to this very limited evidence, are higher than those made by several producers of washed sand and gravel.

243. As might be expected in an extractive industry, the trading results show that both among members of the Associations, and among producers as a whole, there are wide differences in the rates of profit, and this position is not altered to any appreciable extent if account is taken of the alternative figures suggested by the Associations. All producers made profits and, apart from one concern (that which was responsible for about 45 per cent. of all Association sales of sand and gravel and without whose production market requirements would, in all probability, not have been met), the rates of profit on capital employed in 1953 (the best year of those we examined) were high. Our examination shows that there is no clear relationship between the size of turnover and the rate of profit of the various producers. The costs of the individual producers are necessarily affected by the advantages and limitations of the land worked, the proximity of markets for their products, the incidence of overheads, and the level of royalties paid under the leases of the land being operated.

CHAPTER 5: VIEWS OF USERS

244. The principal users of sand and gravel in the Reference Area are public works contractors and builders, and evidence was obtained from a number of these. Public and local authorities, most of them being employers of contractors and builders as well as being in some cases themselves direct users of sand and gravel, also provided evidence. In addition, some builders' merchants participating in the trade were invited to give their views. The principal users of sand and gravel from whom we obtained evidence are among those listed in Appendix M.

The Nature and Pattern of the Demand

245. Most witnesses were concerned with sand and gravel as aggregates in building and civil engineering. The significance of sand and gravel as an element in the cost of these operations varies greatly according to the type of construction (e.g. whether it is in concrete or brick) and, if concrete is used, according to the nature of the aggregate (e.g. whether gravel or crushed rock is used as the coarse aggregate). These variations were illustrated in the evidence of several witnesses.

246. The North of Scotland Hydro-Electric Board estimated that the cost of all aggregates made up about 20 per cent. of the total cost of a dam; of this, "bought-in" sand (mainly washed sand) represented about a quarter. A Glasgow builder said that in a concrete block of flats the cost of aggregates might be 10 to 15 per cent. of the total cost of the building, or more. One large contractor said that sand and gravel accounted on an average for one and a quarter per cent. of the total cost of all his contracts. The Scottish Branch of the Royal Institution of Chartered Surveyors estimated

that, in the case of relatively small general builders, one and a quarter per cent. of the total cost of their building operations was represented by sand and one and a quarter per cent. by gravel; for builders using more concrete the proportions of costs represented by sand and gravel might well be somewhat higher. These cost proportions vary because they refer to different types of construction, with varying types of material, but they suffice to illustrate that sand and gravel is a comparatively small element in most building costs.

247. Users whose work involved the production of high grade concrete said that they were buying an increasingly large proportion of washed, as opposed to unwashed, material. In fact, most witnesses agreed that where sand and gravel were being used for the production of high quality concrete it was necessary that this material should be washed as well as graded. We were told by the Scottish Section of the Federation of Civil Engineering Contractors that since 1939 there had been a big increase in the use of washed sand for concrete making in Central Scotland. Both unwashed screened and unwashed unscreened sand continued to be used for the production of lower grade concrete, but relative demand for these two types had decreased owing to the increased use of washed sand. This change arose from developments in design, requiring the production of concrete with a high ultimate strength, rather than from changes in the availability of different types of sand. In two cases in the evidence we obtained, washed sand was being used for building purposes (e.g. mortar) as well as for concrete work; this was attributed mainly to the poor quality of the sand and gravel deposits available.

Availability and Quality of Supplies

248. One large contractor said that normally only the Association members and one independent producer could supply material of the quality and in the quantity which he required. (Over 70 per cent. of the total supply of washed sand and gravel in the Area is controlled by the Washed Sand and Gravel Association and the Mid-Scotland Washed Sand and Gravel Association; see Table C in paragraph 185 and paragraphs 187-8.) Witnesses were agreed that the sand and gravel deposits in the Area were of very variable quality. Some seemed to think that British Standard Specifications were a counsel of perfection by no means always attainable. Several users stated that they had difficulty in obtaining supplies regularly and consistently to the B.S. Specifications. Important users whose work required high-grade concrete claimed, however, that they consistently bought to B.S. Specifications and the Hydro-Electric Board said that the variations in quality in their supplies lay "on the whole . . . within the limits defined by" these specifications.

249. A number of users did not think washed material necessary for all their operations and preferred to buy unwashed sand. A builder using mainly unwashed material stated that:

"Very often specifications in tenders quote British Standards for sand and aggregates but owing to the lack of proper washing plant at the pits, these are often ignored, the price of importing sand which would comply with B.S.S. being prohibitive".

One witness said that the washing of sand and gravel was a practice introduced from England and that, though washing was necessary for the production of high-grade concrete, it was "a fad or a fashion" for low-grade concrete; he added that in any case not all humus could be removed by washing. We were told by a builder (said to be typical of firms carrying out

all types of building work on a moderate scale) that he usually used unscreened, unwashed sand—including some used for making concrete. Most users, however, considered washing desirable, especially in view of the high proportion of dirt in sand and gravel deposits in the Area. They were agreed that the quality of the washed material supplied in the Area had improved over the last few years; there were, nevertheless, complaints from two bodies representative of users that the water content in washed material was excessive.

Association Arrangements

250. Although nearly all the witnesses buying sand and gravel in the Area mentioned the control over a large part of the supply of washed material by the Associations, apparently none of them had detailed knowledge of Association arrangements and most were unaware of, or were not clear about, the distinction between the West Association and the Mid-Scotland Association; they were also uncertain who were the producer members and who were the merchants of the Associations; one witness was under the impression that an important non-Association producer was in fact a member of the West Association. Most witnesses did, however, know that the Associations fixed prices and operated a central invoicing system.

251. Many users were critical of the uniform minimum price system adopted by the Associations (as distinct from the actual level of prices). Several witnesses said that the Associations were not prepared to reduce their prices in any way, e.g. by discounts for quantity or good custom. (They were naturally unaware of the special prices given to a public and to a local authority, mentioned in paragraph 97.)

252. No specific complaints were made against the Associations' central invoicing system although the Hydro-Electric Board argued that the system played a key part in the West Association's price maintenance. Two merchants gave the obligation to use this system as their reason for not wishing to become Association merchants.

Prices and Price Changes

253. Witnesses provided a good deal of information about sand and gravel prices, including price increases, in the Area. From this it appeared that the Associations' maintained prices for washed materials were generally, though not invariably, somewhat higher than the prices charged by independent producers in the Area. It was also clear that (as has been mentioned in paragraph 208 above) prices in general had risen steadily in recent years. Only two witnesses, however, Perth County Council and the North of Scotland Hydro-Electric Board, expressed active concern at the actual level of Association prices, and only two, Perth County Council and Fife County Council, at non-Association prices; few of the contractors who submitted evidence seemed to have questioned in any detail either the current prices paid by them for material or the increases in these prices, or ever to have investigated the reasons for the increases.

254. Lanark County Council provided us with a list of prices paid for unwashed sand, from which it appeared that ex-pit supplies from a quarry producing unwashed sand but operated by the subsidiary of an Association member were more expensive (by 6d. to 1s. 3d. a ton) than ex-pit supplies of unwashed sand from non-Association producers. Their representatives could not state a reason for this and said that they thought that the price increases for unwashed sand over the last four years had in fact been reasonable; they declined to express an opinion about the increases in the prices of washed materials. The Glasgow Corporation submitted to us a

list of prices paid by them for both washed and unwashed materials ; from this it appeared that the price per ton of unwashed sand had increased by about 2s. since 1950. Their representatives told us that the increases seemed to them fairly to reflect rises in the costs of labour, transport, etc.

255. The contractors who gave evidence offered various explanations for the rises in prices over the last few years. One firm of contractors thought that the rises were due to increased labour and haulage charges but were not prepared to say how far in their view the rises accurately reflected the producer's rising costs. In the opinion of another contractor, prices had increased generally by only about 8 per cent. between 1950 and 1954 and, in the case of gravel, prices quoted varied according to the availability of whinstone as an alternative. Two firms considered that prices in the Area were higher than in other parts of the United Kingdom, one saying that this was because the prices were maintained by the Associations. In general, contractors agreed that the prices of independent producers were lower than those of Association producers but in some cases preference was expressed for Association producers on the grounds that they were better placed to give supplies in large quantities and of good quality.

256. Several witnesses said that gravel prices kept whinstone prices down and that the non-Association producers of sand and gravel seemed to know, and keep just under, the Association prices. An Association of builders considered that the prices charged for washing sand and gravel were in themselves too high, but they did not produce any detailed information to support this contention.

Terms : Water Content and Other Allowances and Conditions of Sale

257. A major complaint among users was the amount of the water content allowance given by the West Association. This allowance was increased on 1st March, 1954, from 5 per cent. on washed sand to $7\frac{1}{2}$ per cent. ; a 2 per cent. allowance was then introduced for certain grades of washed gravel and from June, 1954, a 4 per cent. allowance has been given on certain grades of washed ballast (see notes on Table E in paragraph 207 above and paragraph 209). Some of the witnesses may have been unaware of these changes at the time they gave evidence. Various estimates were made as to the actual water content commonly to be found in washed sand. The Hydro-Electric Board thought that in washed sand on delivery it varied between 8 per cent. and 13 per cent., and estimated that the natural water content in "fairly dry" sand might be 5 per cent. Other witnesses thought that in exceptional cases the water content of washed sand might be considerably higher. The Hydro-Electric Board, the Scottish National Building Trades' Federation, and one contractor thought the allowance still inadequate ; another contractor thought that it was "approaching a fair allowance now", and other users seemed to find the new allowance satisfactory. The evidence of the Associations was that in practice the water content of sand varied very greatly and therefore an arbitrary figure had to be chosen.

258. Bound up with the question of the water content allowance was the resentment expressed by a number of users at the insistence of suppliers in both Associations on the acceptance by customers of weights as recorded at the quarry (i.e. inclusive of much of the water content introduced by washing). This condition is stipulated on the invoices of both Associations. The Associations have told us that in practice they have always been prepared to accept weights taken at weighbridges on the construction site or en route there from the pit ; most of the users from whom we obtained evidence seemed unaware, however, of this concession.